

# build\_workable\_items\_from\_source

## build\_workable\_items\_from\_source.sh

**Revision:** 1 **Last modified:** 2026-09-26T13:00:00Z

### Overview

scripts/pre\_build/build\_workable\_items\_from\_source.sh builds the workable-items binary from its **current** Go source and prints the path of the freshly built file. It never returns a binary that already exists on disk. Pre-build invariant 17 (CM-WORKABLE-ITEMS-VALIDATE) resolves its binary through this script.

### Why this exists

Invariant 17 used to resolve its binary through a candidate loop whose first entry was the git-tracked constitution/scripts/workable-items/bin/workable-items. A tracked binary wins that loop on every fresh clone, so the gate ran whatever an older build could see. On 2026-08-25 the shipped binary was six days behind its source and lacked the terminal-status guards the source had, so ten mis-located tracker rows passed the gate (BOB-188).

The operator decision recorded on BOB-188 (2026-08-26) is **build on demand**: build from source, or refuse when the Go toolchain is absent. Because this script builds fresh on every call, a stale binary cannot be selected at all.

## Prerequisites

- Go on PATH (the same toolchain the qBitTorrent-go backend already needs).
- A C compiler for cgo (the tool links [github.com/mattn/go-sqlite3](https://github.com/mattn/go-sqlite3)).
- The constitution submodule checked out.

## Usage

```
bin="$(bash scripts/pre_build/  
    build_workable_items_from_source.sh)" || rc=$?  
"$bin" validate --db docs/workable_items.db  
rm -rf "$(dirname "$bin")"      # the caller owns cleanup
```

Override the source directory (used by the test fixture):

```
BOBA_WI_SRC_DIR=/path/to/workable-items bash scripts/pre_build/  
    build_workable_items_from_source.sh
```

## Exit codes

| Code | Meaning                                                        |
|------|----------------------------------------------------------------|
| 0    | Built; the absolute path of the binary is on stdout            |
| 1    | go build failed — no path printed                              |
| 2    | Source directory missing, or it has no cmd/workable-items      |
| 3    | REFUSED — Go is not on PATH; no binary on disk is used instead |

## Edge cases

- A compile error never falls back to an older binary: exit 1 with nothing on stdout.
- A missing toolchain is a refusal (exit 3), and invariant 17 turns it into a FAIL, not a silent SKIP.
- The binary lives in a `mktemp -d` directory named `boba-wi-build.*`; invariant 17 removes only a directory matching that name.

## Internal behaviour

Resolve the source directory, check for go, go build -o <tmpdir>/workable-items ./cmd/workable-items, verify the output is an executable file, print its path. A warm Go build cache makes a rebuild take about two seconds (measured 2026-09-26).

## Related scripts

- scripts/pre\_build/check\_cm\_workable\_items\_binary\_fresh.sh — invariant 52, fingerprint check on the shipped binary, kept as defence-in-depth for as long as a shipped binary exists on disk.
- tests/pre\_build/test\_build\_workable\_items\_from\_source.sh — RED/GREEN cases, a negative control that replays the old candidate loop, and a paired mutation.

## Last verified

2026-09-26 — bash tests/pre\_build/test\_build\_workable\_items\_from\_source.sh reported 6 passed, 0 failed.